DEPC-treated water (nuclease-free water for RNA and DNA work)

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v1.0, 2026-10-21; 0.1% DEPC, overnight, autoclaved

Purpose: water free of RNases (and DNases) for elution buffers, PCR set-up, RNAlater and library work. DEPC (diethyl pyrocarbonate) inactivates nucleases by carbethoxylating histidine residues; autoclaving then destroys the remaining DEPC (it decomposes into CO_2 and ethanol).

Reagent: diethyl pyrocarbonate, molecular biology grade (e.g. HiMedia MB076, himedialabs.com/us/mb076-depc.html). Store the DEPC bottle at 2–8 °C, tightly closed, away from moisture; open it only in the fume hood.

Safety: DEPC is volatile, a suspected carcinogen and reacts violently with ammonia and amines. Fume hood, nitrile gloves, lab coat, safety glasses. Never add DEPC to buffers that contain Tris, HEPES or any other amine: treat the water first, then dissolve the buffer salts in the treated, autoclaved water.

Procedure

TREATMENT

1. In the fume hood, add **1 mL** DEPC to **1 L** Milli-Q (or double-distilled) water in a clean glass bottle (final **0.1% v/v**). Use a bottle with a loose cap or foil; never a sealed vessel.

2. Shake or stir vigorously until the DEPC droplets have dispersed (DEPC is poorly soluble), then stir **≥ 1 h** more.

3. Incubate **overnight (≥ 12 h) at 37 °C** (room temperature works with a longer incubation). Keep the bottle in the hood or a ventilated incubator: DEPC vapour escapes during this time.

INACTIVATION

4. Autoclave **121 °C, 20 min** with the cap loosened. This hydrolyses the remaining DEPC; the water must smell of nothing afterwards. A sweet, fruity smell means DEPC is still present: autoclave again.

Note Un-destroyed DEPC carbethoxylates adenine residues and will inhibit downstream enzymes. Do not skip or shorten the autoclave step.

5. Let the bottle cool with the cap loosened, then close it tightly. Label it “DEPC water”, date, initials.

USE AND STORAGE

6. Store at room temperature, tightly closed. Open only with nuclease-free technique (gloves, clean bench, never pipette back into the bottle). Aliquot into 50 mL nuclease-free tubes for daily use and discard an aliquot once it has been opened for more than a month.

7. Buffers: dissolve the salts in DEPC-treated, autoclaved water and autoclave the buffer again. Do not treat the finished buffer with DEPC.

Note Commercial nuclease-free water is an acceptable substitute for small volumes; DEPC water is the lab's stock for elution, SD-buffer dilution and RNAlater preparation.

Materials

#	Material
1	Diethyl pyrocarbonate, molecular biology grade (HiMedia MB076 or equivalent)
2	Milli-Q or double-distilled water
3	1 L glass media bottle (autoclavable) with loosened cap or foil
4	Magnetic stirrer and bar (or shaker)
5	**37 °C incubator; autoclave**
6	Fume hood, nitrile gloves, safety glasses